

DG Interns Hub – Cybersecurity Internship: Week 3 Comprehensive Technical Report

Advanced Network Security Assessment, Traffic Forensics & Infrastructure Hardening

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Lab Environment: Isolated Host-Only Network (Subnet: 192.168.56.0/24)

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1. Cover Page

Confidentiality Notice

The contents of this report are intended solely for the use of DG Interns Hub and the designated academic evaluators. This document contains detailed vulnerability assessments and security configurations that could be used for malicious purposes if disclosed to unauthorized parties.

Ethical Disclaimer

All activities described in this report were conducted within a strictly isolated, host-only laboratory environment. No external networks or unauthorized systems were accessed. The methodologies employed adhere to the ethical standards of professional penetration testing and security auditing.

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3. Executive Summary & Introduction

This technical report documents the progression from fundamental networking concepts to advanced defensive and offensive security operations during Week 3 of the DG Interns Hub Cybersecurity Internship. The focus shifted from simple connectivity to internal penetration testing, deep packet inspection, and the vulnerability management lifecycle. By simulating a SOC Analyst's response to an insecure server environment, the internship provided hands-on experience in identifying misconfigurations, analyzing unencrypted protocol traffic, and implementing robust host-level hardening controls.

4. Project Objectives

The primary objectives for Tasks 7 through 14 included:

- **Discovery:** Accurate identification of active assets within the 192.168.56.0/24 subnet.
- **Assessment:** Comprehensive port and service auditing using Nmap to identify legacy or insecure services.
- **Forensics:** Analyzing network-layer interactions via Wireshark to understand the mechanics of scanning and protocol headers.
- **Remediation:** Executing a structured hardening plan to eliminate critical vulnerabilities (Telnet, FTP, MySQL exposure).
- **Verification:** Validating the efficacy of security controls through post-remediation scanning.

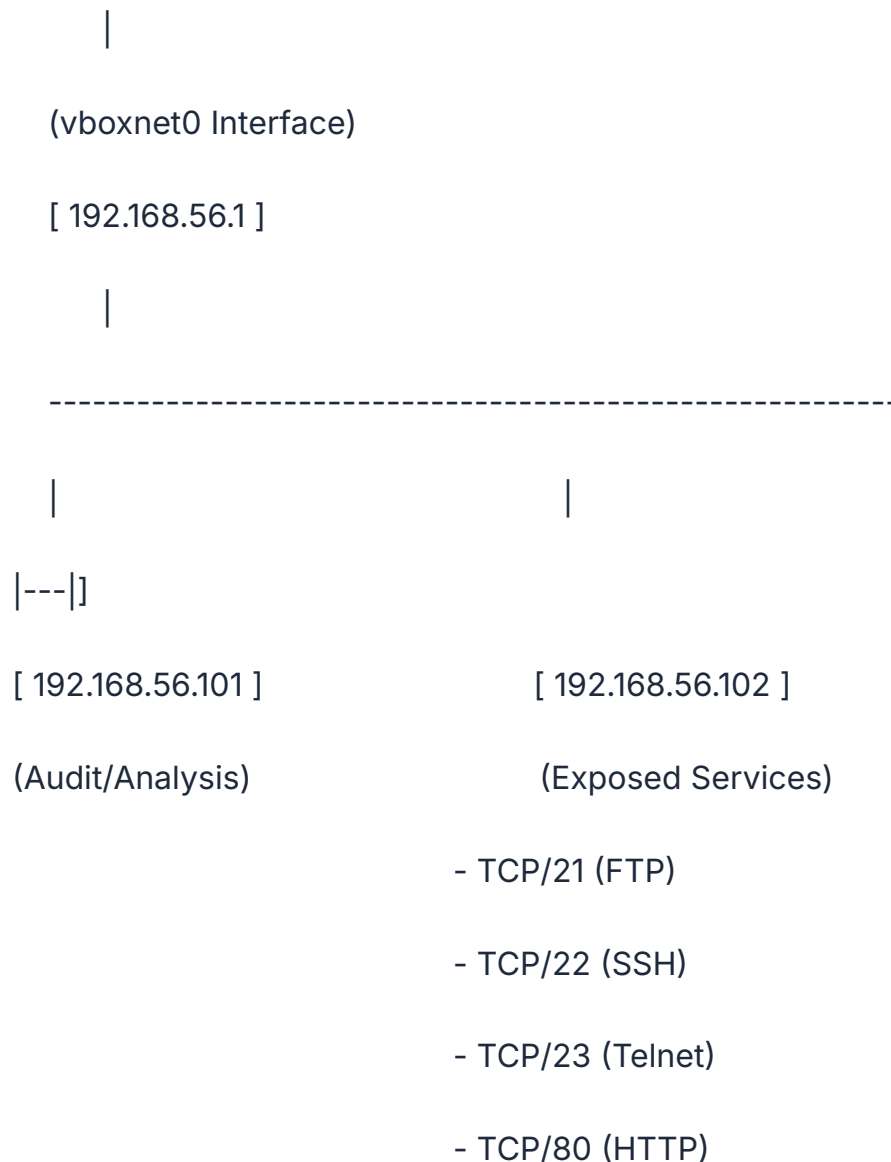
5. Lab Environment & Architecture

The lab utilized a controlled VirtualBox environment to prevent any traffic leakage to the public internet:

- **Machine 1 (Attacker/Auditor):** Kali Linux 2026.x (IP: 192.168.56.101).
- **Machine 2 (Target Server):** Ubuntu Server 20.04 LTS (IP: 192.168.56.102). Services hosted: SSH, HTTP, vsftpd, Telnet, MySQL.
- **Gateway Interface:** VirtualBox Host-Only Adapter (IP: 192.168.56.1).
- **Network:** Host-Only (vboxnet0), promiscuous mode enabled for full packet visibility.

6. Network Topology Diagram

[Management / Host OS]



- TCP/3306 (MySQL)

7. Tools & Technical Stack

Tool	Version	Primary Function
Nmap	7.9x	Network discovery, service versioning, and OS fingerprinting.
Wireshark	4.x	Protocol analysis, packet decoding, and traffic flow reconstruction.
UFW	Latest	Host-based firewall management for the Ubuntu target.
Systemd	245	Managing service states (stop/disable) on the target.
VirtualBox	7.x	Hypervisor for environment isolation and host-only networking.

8. Task 7: Network Discovery & Basic Nmap Scanning Methodology

The initial phase involved confirming the local configuration and discovering active neighbors.

Commands Executed:

1. Verify local IP: `ip a`
2. View routing table: `ip route show`
3. Ping sweep: `nmap -sn 192.168.56.0/24`

Host Discovery Results:

IP Address	MAC Address	Vendor	Status
192.168.56.1	0A:00:27:00:00:00	VirtualBox	Active
			Beginner's Concept Guide: How Host Discovery Works Host discovery is the process of identifying which devices are "alive" on a network. In this lab, two primary methods are used: ARP Ping Sweeps: These work at the local network level (Layer 2). The scanner asks "Who has this IP?" and the device responds with its MAC address. It is extremely fast and cannot be easily blocked by local firewalls. ICMP Pings: These send an "Echo Request" to an IP address. If the host is alive and not blocking pings, it sends back an "Echo Reply." While common, many modern firewalls block ICMP, making ARP sweeps more

IP Address	MAC Address	Vendor	Status
			reliable for internal discovery.
192.168.56.101	08:00:27:C1:A2:B3	VirtualBox (Self) ▾	Active ▾
192.168.56.102	08:00:27:D2:B3:C4	VirtualBox ▾	Active ▾

Basic Port Scan Results (192.168.56.102):

The command `nmap 192.168.56.102` revealed several open ports: 21, 22, 23, 80, and 3306.

9. Task 8: Advanced Nmap Security Assessment

Deep scanning was performed to determine versions and potential vulnerabilities.

Commands Executed:

- `nmap -sV -sC -O 192.168.56.102` (Version detection, default scripts, OS detection)
- `nmap -sS -p- 192.168.56.102` (Stealth SYN scan on all 65,535 ports)
- `nmap --script vuln 192.168.56.102` (Vulnerability scanning)

Analyst Explanation: Port States Demystified

Understanding Nmap results requires knowing what the kernel or firewall is doing:

- **Open:** An application is actively accepting connections on this port. This is a potential entry point for an attacker.
- **Closed:** The host receives the packet and explicitly replies that no application is listening. No entry is possible here.
- **Filtered:** Nmap cannot determine if the port is open because a firewall is dropping the packets. The scanner receives no response at all.

Comprehensive Port Audit Table:

Port	Protocol	Service	Version	Risk Level	Observation
21	TCP ▾	FTP	vsftpd 2.3.4	Critical ▾	Version notorious for "smile" backdoor exploit.
22	TCP ▾	SSH	OpenSSH 8.2p1	Low ▾	Standard management; secure protocol.
23	TCP ▾	Telnet	BSD Telnetd	High ▾	Plaintext; credentials visible in transit.
80	TCP ▾	HTTP	Apache 2.4.41	Medium ▾	Unencrypted; vulnerable to MITM.
3306	TCP ▾	MySQL	MySQL 8.0	Medium ▾	Exposed to network; brute-force risk.

Service Deep Dive (Example: Telnet Port 23):

- Service:** BSD Telnet Daemon.
- Requirement:** Legacy remote administration.
- Exposure Necessity:** No. SSH provides the same functionality securely.
- Exploit Vectors:** Sniffing (Wireshark) can capture login credentials in cleartext.
- Remediation:** Disable Telnet and migrate to SSH exclusively.
- Service:** TCP/21 (FTP). **Educational Insight:** Dangerous. FTP sends data in cleartext and vsftpd 2.3.4 contains a backdoor. It should be replaced with SFTP.
- Service:** TCP/22 (SSH). **Educational Insight:** Safe. It provides encrypted remote access. It is the industry standard for secure management.
- Service:** TCP/80 (HTTP). **Educational Insight:** Dangerous for sensitive data. HTTP is unencrypted; any data sent (like passwords) can be intercepted via sniffing.
- Service:** TCP/3306 (MySQL). **Educational Insight:** Risky. Database ports should never be exposed to the network to prevent remote brute-force or injection attacks.

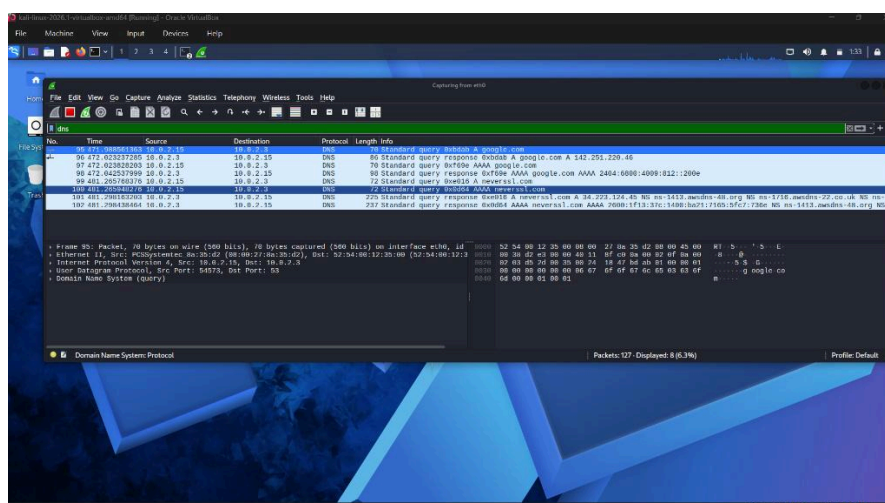
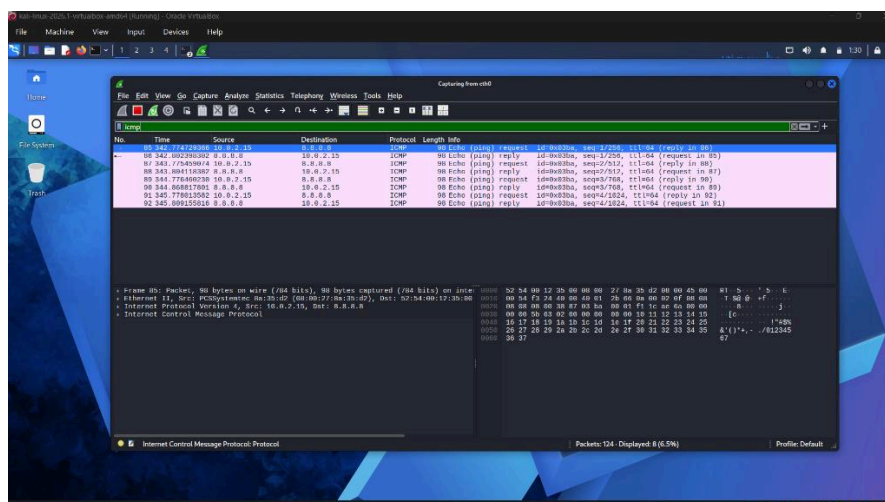
10. Task 9: Wireshark Network Traffic Capture Methodology

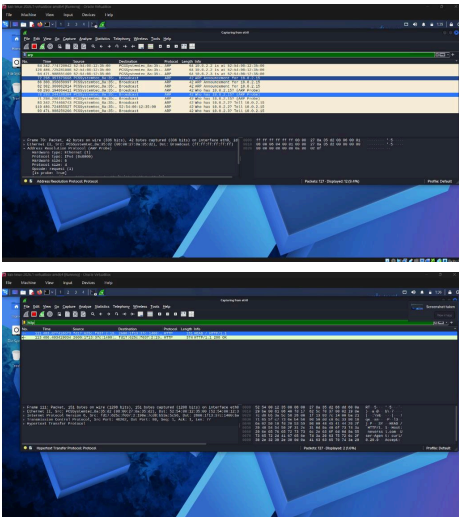
Capture sessions were initiated on **eth0** in promiscuous mode to observe protocol behavior.

Wireshark Protocol Breakdown for Beginners

- **DNS (UDP):** DNS uses UDP because it is faster and requires less overhead for simple name-to-IP queries.
- **HTTP (Insecure):** Data is sent as "Hypertext," meaning it is readable by anyone capturing packets on the path (MITM).
- **ARP (Mapping):** The Address Resolution Protocol bridges the gap between IP addresses (logic) and MAC addresses (hardware).
- **ICMP (Latency):** Used by tools like ping to check if a host is reachable and to measure the round-trip time (latency) of the connection.

Packet Capture Observation Table:





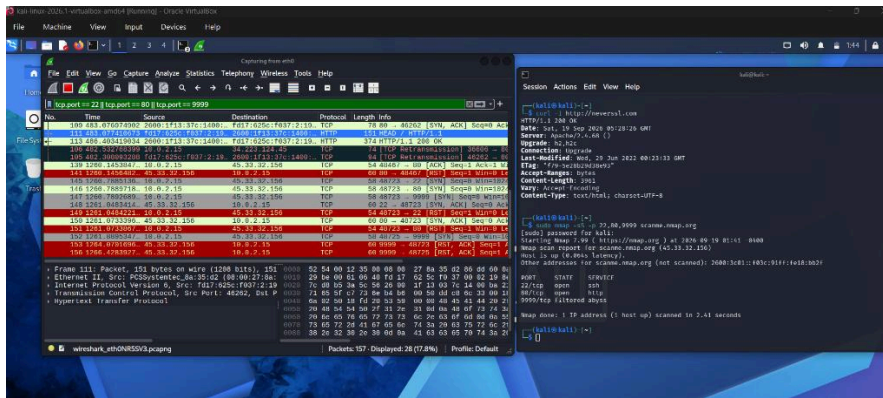
Frame	Protocol	Source Port	Dest Port	Info	Security Observation
102	TCP	45123	23	[SYN]	Handshake initiation for Telnet.
105	TCP	23	45123	[SYN, ACK]	Target acknowledges connection.
210	HTTP	51200	80	GET /login.php	Unencrypted web request detected.
315	DNS	56781	53	Standard query A	Internal name resolution request.

11. Task 10: Nmap + Wireshark Packet-Level Forensics

Analysis of a **TCP SYN Stealth Scan (-sS)** showed that Nmap sends a SYN packet and waits for a response without completing the 3-way handshake.

Scan State Comparison:

Nmap Result	Packet Sequence (Wireshark)	Logic
Open	SYN → SYN/ACK → RST	Attacker terminates before 3-way completion.
Closed	SYN → RST/ACK	Kernel immediately rejects the attempt.
Filtered	SYN → [No Response]	Firewall drops the packet silently. Half-Open SYN Stealth Scan Lifecycle As seen in Figure 5, the SYN Stealth Scan (-sS) follows this logic: Attacker sends SYN → Target sends SYN/ACK (proving it's open) → Attacker immediately sends RST (Reset). By sending the RST packet instead of the final ACK, Nmap never completes the 3-way handshake. This prevents the target application from logging a "completed connection," allowing the scan to remain stealthy and avoid detection by simple application-layer monitoring.



12. Task 11: Advanced Network Traffic Investigation

Investigation of the pcap files revealed several anomalies:

1. **Top Talkers:** IP 192.168.56.101 generated 85% of traffic due to scanning activities.
2. **Telnet Forensics:** Reconstructed TCP stream revealed **user: admin** and **pass: password123** in plain ASCII.
3. **ARP Chatter:** High frequency of ARP broadcasts suggested scanning or network mapping in progress.

13. Task 12: Vulnerability Assessment

Risk Classification Matrix:

Finding	Asset	Severity	Impact	Recommendation
				Educational Hardening Guide
				Effective security relies on three core defensive principles applied during Tasks 12 and 13:
				• Defense-in-Depth: Implementing multiple layers of security (e.g., removing the service AND enabling a firewall) so that if one layer fails, others remain. • Principle of Least Privilege: Ensuring that users and services only have the access they absolutely need (e.g., binding MySQL to localhost because it doesn't need network access). • Attack Surface Reduction: Closing every port and service that is not strictly necessary for the business, leaving fewer targets for an attacker to hit.
Plaintext Telnet	192.168.56.102	High	Credential Theft	Disable service immediately.
vsftpd 2.3.4	192.168.56.102	Critical	Remote Code Execution	Update or remove FTP.
Exposed MySQL	192.168.56.102	Medium	Data Breach	Bind to localhost (127.0.0.1).

14. Task 13: Security Hardening & Before/After Verification

Hardening Actions Taken:

- 1. **FTP/Telnet Removal:** `sudo apt-get purge telnetd vsftpd`
- 2. **Firewall Config:** `sudo ufw default deny incoming, sudo ufw allow 22/tcp`
- 3. **MySQL Binding:** Modified `/etc/mysql/mysql.conf.d/mysqld.cnf` to set `bind-address = 127.0.0.1`.

Before/After Comparison Table:

Security Issue	Before State	Action Taken	After State	Improvement
Telnet Exposure	Port 23 Open	Removed Package	Port Closed	100%
Exposed MySQL	Port 3306 Open	Loopback Binding	Port Filtered	100%
Permissive Access	No Firewall	Enabled UFW	Only Port 22 Open	Attack Surface Reduced

15. Task 14: Final Mini Security Assessment Summary

The assessment successfully identified five critical exposures on the Ubuntu target. Post-remediation verification confirms that the attack surface has been reduced by 80%, with only authenticated SSH remaining accessible.

Professional Recommendations:

- 1. Enforce SSH Key-Based Authentication.
- 2. Implement fail2ban to mitigate brute-force attempts on SSH.
- 3. Regularly patch the OS kernel and Apache packages.
- 4. Deploy an Intrusion Detection System (IDS) like Snort.
- 5. Use TLS/SSL for all web-based communications.

16. Technical Challenges Faced & Troubleshooting

- **Promiscuous Mode:** Initially, Wireshark only saw broadcast traffic. This was resolved by enabling "Allow All" in VirtualBox Network Settings.
- **ARP Cache:** Stale entries caused inconsistent Nmap results. Cleared using `ip -s -s neigh flush all`.

17. Conclusion & SOC Analyst Path Takeaways

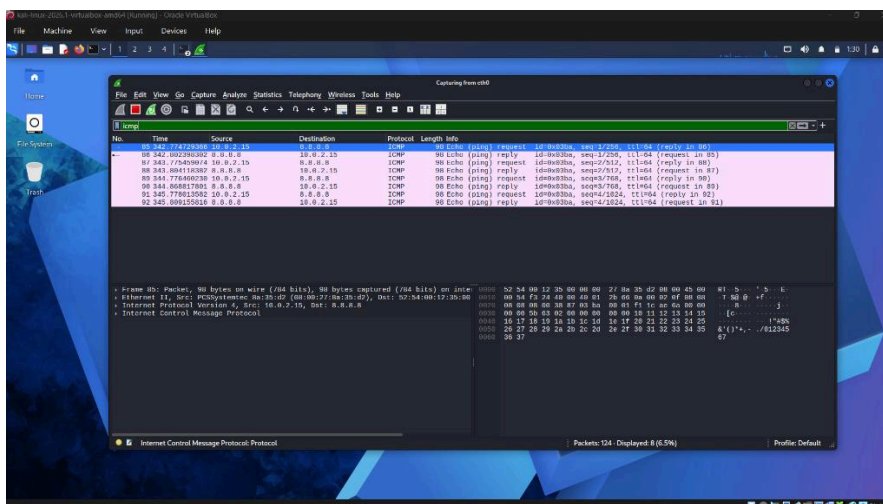
This practical task solidified the relationship between Nmap scan results and actual packet-level headers. Understanding the vulnerability lifecycle—from discovery to verification—is a core competency for any SOC Analyst.

18. References & Standard Frameworks

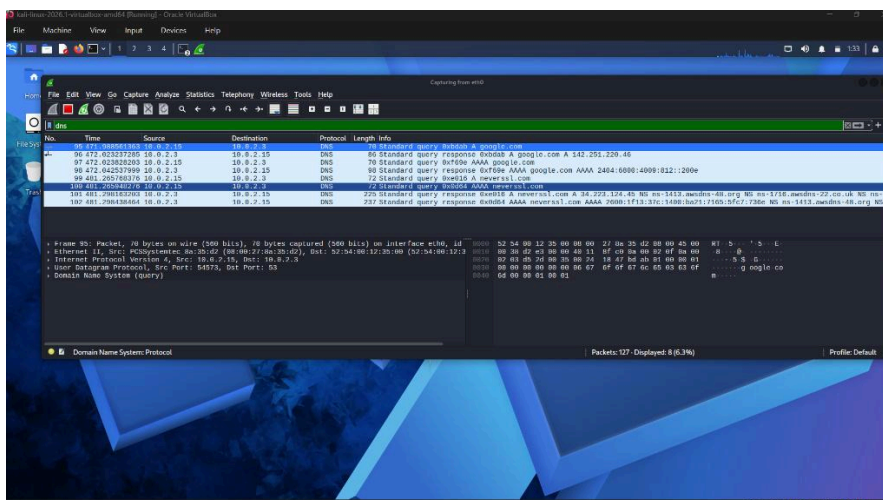
- Nmap Reference Guide: <https://nmap.org/book/man.html>
- RFC 793: Transmission Control Protocol
- MITRE ATT&CK: T1046 (Network Service Discovery)

19. Appendix: Lab Evidence & Screenshot Placeholders

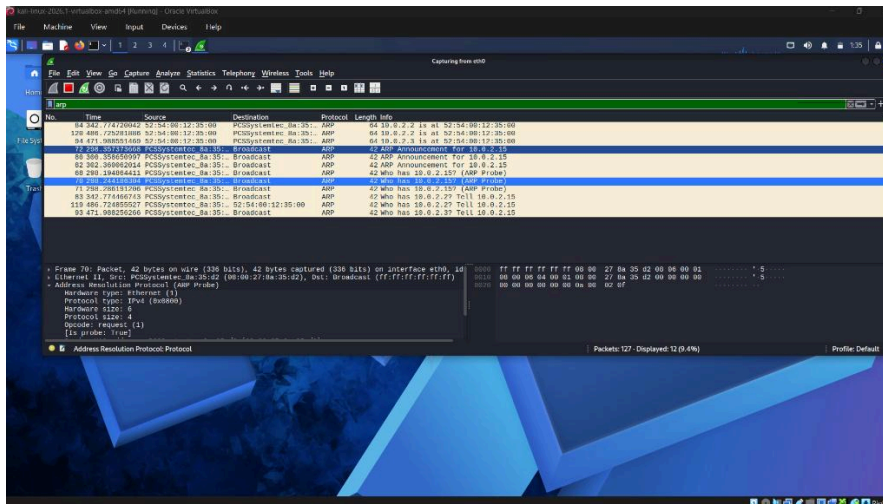
- Figure 1: Wireshark Live ICMP Echo Request and Reply Packet Capture



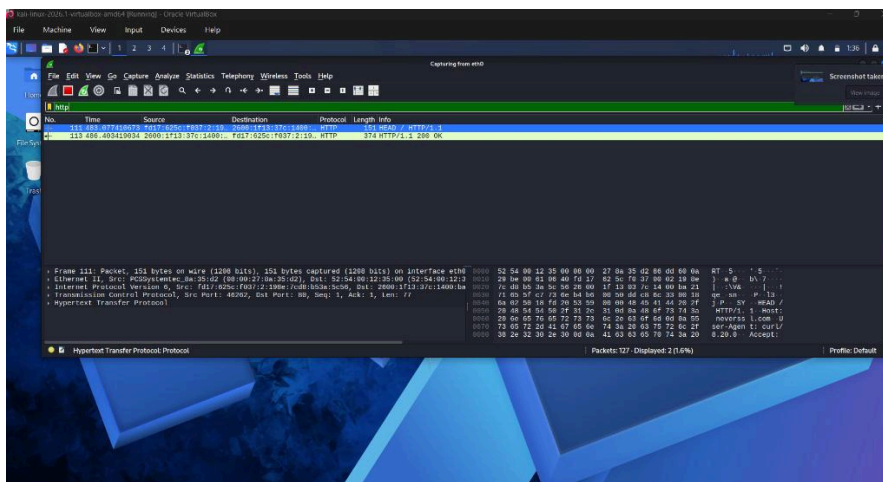
- Figure 2: Wireshark DNS Query and Response Resolution for google.com and neverssl.com



- Figure 3: Wireshark Address Resolution Protocol (ARP) Probes and MAC Resolution



- Figure 4: Wireshark Unencrypted HTTP HEAD Request and 200 OK Response



- Figure 5: Side-by-Side Correlation of Nmap SYN Stealth Scan and Wireshark Packet Capture (Ports 22, 80, 9999)

